Motor Thrust Stand



Category: **Drones & Aviation** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Motor Thrust Stand; Thrust Measuring Stand

What it is used for

A motor thrust stand measures the thrust produced by an electric motor and propeller or other test assembly. Secure mounting, correct instrumentation, controlled power application, and effective guarding are essential because rotating propellers and high-current electrical systems can cause serious injury.

Main parts

- Rigid frame supports the motor and test assembly.
- Motor mount secures the motor in the specified orientation.
- Load cell measures the force produced by the motor.
- Signal conditioner or amplifier processes the load-cell output where fitted.
- Electronic speed controller regulates motor speed where applicable.
- Power supply or battery provides electrical energy.
- Voltage and current sensors measure electrical input where fitted.
- Data acquisition unit records thrust, speed, voltage, and current.
- Propeller guard or test enclosure limits access to the rotating assembly where fitted.

Before you start

1. Only trained personnel should operate the thrust stand and high-power motor test equipment.
2. Inspect the frame, motor mount, fasteners, load cell, wiring, connectors, and test enclosure for damage.
3. Confirm the motor, propeller, ESC, battery, and stand are compatible and within their rated limits.
4. Secure the stand to a stable test bench and verify that the motor mount cannot shift.
5. Balance and inspect the propeller according to its instructions; do not use cracked, chipped, or damaged propellers.
6. Calibrate or zero the load cell and verify the sensor and data acquisition connections.
7. Clear the test zone of people and loose objects, and use appropriate eye protection and hearing protection.

How to use

1. Mount the motor securely and install the specified propeller only when the power source is disconnected.
2. Connect the ESC, power supply or battery, and sensors according to the test wiring diagram.
3. Check polarity, connector security, sensor zero, and data recording before energising the system.

4. Fit the required propeller guard or enclosure and verify that all personnel are outside the exclusion zone.
5. Start the data acquisition system and confirm that thrust, speed, voltage, and current readings are plausible.
6. Energise the motor and increase throttle gradually according to the approved test plan.
7. Record readings at the specified operating points without exceeding motor, propeller, ESC, battery, or stand limits.
8. Monitor vibration, temperature, current, thrust readings, and unusual sound throughout the test.
9. Reduce throttle to zero and disconnect the power source before approaching the motor or propeller.
10. Wait until the propeller has completely stopped before removing guards, adjusting sensors, or changing components.
11. Save the test data, inspect the assembly for damage, and document any abnormal readings.

⚠ Safety

- Never touch or approach a rotating propeller, even at low throttle.
- Keep people outside the propeller plane and designated exclusion zone during testing.
- Never exceed the rated voltage, current, speed, or thrust of any component.
- Disconnect the battery or power source before changing wiring, mounting, or propeller components.
- Do not operate with a cracked propeller, loose motor mount, damaged wiring, or unprotected rotating parts.
- Stop the test immediately if severe vibration, unexpected movement, smoke, overheating, or sensor failure occurs.

Care & storage

- Remove power before cleaning or inspecting the stand.
- Inspect the load cell, mounts, fasteners, connectors, guards, and wiring before each test session.
- Keep the load cell free from side loading and impact beyond its rated limits.
- Check sensor calibration and data acquisition operation at the scheduled interval.
- Store the stand and test accessories in a dry, secure location and replace damaged components.

Common mistakes to avoid

- Testing with an unbalanced, damaged, or incorrectly installed propeller.
- Approaching the motor before the propeller has fully stopped.
- Ignoring load-cell zero errors or unreliable sensor readings.
- Increasing throttle beyond the ratings of the motor, ESC, propeller, or power source.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.